# John Steinman

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Dept. of Computational Applied Mathematics & Operations Research
Rice University
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## Education

**Ph.D., Computational Applied Mathematics and Operations Research** (expected) May 2027
Rice University
Advisor: Matthias Heinkenschloss
GPA: 4.0

**M.A., Computational Applied Mathematics and Operations Research** Dec 2024
Rice University
Advisor: Matthias Heinkenschloss
GPA: 4.0

**B.S., Mathematics and B.S., Computational Engineering** May 2022
The University of Texas at Austin
GPA: 3.98

## Research and Work Experience

**Rice University** Graduate Student Researcher, Aug 2022–Present
*Dept. of Computational Applied Mathematics & Operations Research*
- Researched optimization for problems with surrogate model inexactness

**Sandia National Laboratories,** Summer Intern, Jun 2024–Aug 2024
*Optimization and Uncertainty Quantification*
- Researched preconditioning for spectral collocation methods

**Oden Institute** Undergraduate Student Researcher, Sep 2020–May 2022
*The University of Texas at Austin*
- Developed a computational framework for estimating material parameters in hydrogel media

**Firefly Aerospace, Cedar Park, TX** Summer Intern, Jun 2020–Aug 2022
*Guidance, Navigation, and Control Team*
- Worked on rocket trajectory optimization and flight parameter tuning

**MD Anderson Cancer Center, Houston TX** Summer Intern, Jun 2019–Aug 2019
*Department of Biostatistics*
- Developed statistical models for breast cancer diagnoses based on gene expression data

## Publications

1. Matthias Heinkenschloss and John D. Steinman. Rockafellian relaxation for optimization with model inexactness. Preprint forthcoming, 2026

2. Aurya Javeed, Drew P. Kouri, Denis Ridzal, and John D. Steinman. A preconditioner for spectral collocation. *SIAM Journal on Scientific Computing*, 47(5):A2828–A2850, 2025. doi:10.1137/24M1712539

3. Aurya Javeed, Drew P. Kouri, Denis Ridzal, Isaac M. Ross, and John D. Steinman. Matrix-free linear algebra for trajectory optimization. Submitted to Journal of Guidance, Control, and Dynamics, 2024. URL: https:

//arxiv.org/abs/2509.01855

4. John D. Steinman. Convergence results and a new preconditioner for spectral collocation in time. Master's thesis, Rice University, Houston, TX, December 2024. URL: https://hdl.handle.net/1911/118215

5. Alex Khang, John D. Steinman, Robin Tuscher, Xinzeng Feng, and Michael S. Sacks. Estimation of aortic valve interstitial cell-induced 3D remodeling of poly(ethylene glycol) hydrogel environments using an inverse finite element approach. *Acta Biomaterialia*, 160:123–133, 2023. doi:10.1016/j.actbio.2023.01.043

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## Talks and Presentations

1. John D. Steinman. Rockafellian relaxation for optimization problems with model inexactness. Presentation, Research Training Group in Numerical Mathematics and Scientific Computing at Rice University Annual Workshop, 3 Oct. 2026, Houston, TX, 2026

2. John D. Steinman. Rockafellian relaxation for optimization problems with model inexactness. Presentation, SIAM Conference on Optimization, 5 Jun. 2026, Edinburgh, UK, 2026

3. John D. Steinman. Rockafellian relaxation for optimization problems with model inexactness. Presentation, Research Training Group in Numerical Mathematics and Scientific Computing at Rice University Annual Ranch Retreat, 9 Apr. 2026, Galveston, TX, 2026

4. John D. Steinman. Rockafellian relaxation for optimization problems with model inexactness. Presentation, Research Training Group in Numerical Mathematics and Scientific Computing at Rice University Annual Workshop, 3 Oct. 2025, Houston, TX, 2025

5. John D. Steinman. Matrix-free linear algebra for trajectory optimization. Presentation, 8th Annual Meeting of the SIAM Texas-Louisiana Section, 26 Sept. 2025, Austin, TX, 2025

6. John D. Steinman. A scalable collocation method for trajectory optimization. Presentation, Research Training Group in Numerical Mathematics and Scientific Computing at Rice University Annual Ranch Retreat, 4 May 2025, Houston, TX, 2025

7. John D. Steinman. Matrix-free linear algebra for trajectory optimization. Presentation, Lamar University EXPO 2025 Conference, 23 Apr. 2025, Beaumont, TX, 2025

8. John D. Steinman. Matrix-free linear algebra for trajectory optimization. Presentation, East Coast Optimization Meeting (ECOM), 18 Apr. 2025, Arlington, VA, 2025

9. John D. Steinman. Matrix-free linear algebra for trajectory optimization. Presentation, 2025 SIAM Conference on Computational Science and Engineering (CSE25), 4 Mar. 2025, Fort Worth, TX, 2025

10. John D. Steinman. A preconditioner for spectral collocation. Poster, 7th Annual Meeting of the SIAM Texas-Louisiana Section, 11 Oct. 2024, Waco, TX, 2024

11. John D. Steinman. On the convergence of collocation methods for initial value problems. Presentation, Research Training Group in Numerical Mathematics and Scientific Computing at Rice University Annual Workshop, 8 Oct. 2024, Houston, TX, 2024

12. John D. Steinman. On the convergence of collocation methods for initial value problems. Presentation, Research Training Group in Numerical Mathematics and Scientific Computing at Rice University Annual Ranch Retreat, 20 Apr. 2024, Houston, TX, 2024

13. John D. Steinman. Impact of representation of collocation methods on dynamic optimization problems. Poster, 6th Annual Meeting of the SIAM Texas-Louisiana Section, 4 Nov. 2023, Lafayette, LA, 2023

14. John D. Steinman. Impact of representation of collocation methods on dynamic optimization problems. Poster, Research Training Group in Numerical Mathematics and Scientific Computing at Rice University Annual Workshop, 13 Oct. 2023, Houston, TX, 2023

15. John D. Steinman, A. Khang, X. Feng, and M. S. Sacks. Simulation of the local mechanical behavior of 3D poly(ethylene glycol) hydrogels for studying cell mechanics. Presentation, Annual Gulf Coast Undergraduate Research Symposium, 16 Oct. 2021, Houston, TX, 2021

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## Honors and Awards

**Best Oral Presentation in Graduate STEM** Apr 2025
*Lamar University Spring Expo 2025*

**Ken Kennedy Institute Computational Science and Engineering Recruiting Fellowship** 2022-Present
*Rice University*
- $15,000 awarded over 4 years

**Dr. Hans M. Mark Scholars Endowment in Engineering Honors** 2018-2022
*The University of Texas at Austin*
- $56,000 awarded over 4 years

**H. Bascom Funchess Jr. Scholarship** 2018-2022
*The University of Texas at Austin*
- $12,000 awarded over 4 years

**Distinguished College Scholar** The University of Texas at Austin 2019-2022

**University Honors,** The University of Texas at Austin 2018-2022

**National Merit Scholarship** 2018
- $1,500 award

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## Service

**Rice CMOR Grad Seminar Organizer,** Rice University Aug 2025–Present
- Organized weekly graduate seminars for the CMOR department

**SIAM Student Chapter President,** Rice University Aug 2025–Aug 2026
- Organized SIAM graduate student events and managed chapter finances

**Ranch Retreat Organizer,** Rice University Jan 2024–Apr 2024
*Research Training Group (RTG) in Numerical Mathematics and Scientific Computing*
- Organized first annual RTG Ranch Retreat in Houston, TX
- Invited graduate student and postdoctoral speakers from other universities

**Graduate Recruitment Organizer,** Rice University Feb 2024
*Dept. of Computational Applied Mathematics & Operations Research*
- Organized visit weekends for prospective Ph.D. students

**Academic Coach,** The University of Texas at Austin Jan 2020–Sep 2020
*Ramshorn Scholar Program*
- Mentored freshmen engineering students and provided supplemental course instruction

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## Teaching

**Teaching Assistant,** Rice University Jan 2026–May 2026
*Dept. of Computational Applied Mathematics & Operations Research*
- CMOR 304 (Differential Equations), Spring 2026

**Grader,** Rice University Aug 2022–Jan 2026
*Dept. of Computational Applied Mathematics & Operations Research*
- CMOR 433/533 (Numerical Optimization), Spring 2025

- CMOR 431/531 (Convex Optimization), Fall 2024
- CMOR 421/521 (High Performance Computing), Spring 2024
- CMOR 420/520 (Computational Science), Fall 2023
- CAAM 336 (Differential Equations), Fall 2022, Spring 2023

**Academic Tutor,** The University of Texas at Austin Jan 2020–May 2020
- Calculus, differential equations, physics, chemistry, and other engineering classes

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## Skills

**Programming and Development Tools:** Python, C++, Julia, MATLAB, Git, CMake, Linux, LaTeX
**Optimization and Scientific Computing:** CasADi, IPOPT, ROL, Gurobi, Pyomo, JAX, NumPy, SciPy, FEniCS
**Parallel and High-Performance Computing:** MPI, OpenMP, Slurm

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## References

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